8.3 Homogeneous and universal graphs

Unlike finite graphs, infinite graphs offer the possibility to represent an entire graph property $\mathcal{P}$ by just one specimen, a single graph that contains all the graphs in $\mathcal{P}$ up to some fixed cardinality. Such graphs are called 'universal' for this property.

More precisely, if $\leqslant$ is a graph relation (such as the minor, topological minor, subgraph, or induced subgraph relation up to isomorphism), we call a countable graph G^* *universal* in $\mathcal{P}$ (for $\leqslant$) if $G^* \in \mathcal{P}$ and $G \leqslant G^*$ for every countable graph $G \in \mathcal{P}$.

universal

Is there a graph that is universal in the class of *all* countable graphs? Suppose a graph R has the following property:

Whenever U and W are disjoint finite sets of vertices in R, there exists a vertex $v \in R - U - W$ that is adjacent in R to all the vertices in U but to none in W. $(*)$

Then R is universal even for the strongest of all graph relations, the induced subgraph relation. Indeed, in order to embed a given countable graph G in R we just map its vertices $v_1, v_2, \dots$ to R inductively, making sure that v_n gets mapped to a vertex $v \in R$ adjacent to the images of all the neighbours of v_n in $G[v_1, \dots, v_n]$ but not adjacent to the image of any non-neighbour of v_n in $G[v_1, \dots, v_n]$. Clearly, this map is an isomorphism between G and the subgraph of R induced by its image.

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Theorem 8.3.1. (Erdős & Rényi 1963)
There exists a unique countable graph R with property $(*)$.

Proof. To prove existence, we construct a graph R with property $(*)$ inductively. Let $R_0 := K^1$. For all $n \in \mathbb{N}$, let R_{n+1} be obtained from R_n by adding for every set $U \subseteq V(R_n)$ a new vertex v joined to all the vertices in U but to none outside U. (In particular, the new vertices form an independent set in R_{n+1}.) Clearly $R := \bigcup_{n \in \mathbb{N}} R_n$ has property $(*)$.

To prove uniqueness, let $R = (V, E)$ and $R' = (V', E')$ be two graphs with property $(*)$, each given with a fixed vertex enumeration. We construct a bijection $\varphi\colon V \to V'$ in an infinite sequence of steps, defining $\varphi(v)$ for one new vertex $v \in V$ at each step.

At every odd step we look at the first vertex v in the enumeration of V for which $\varphi(v)$ has not yet been defined. Let U be the set of those of its neighbours u in R for which $\varphi(u)$ has already been defined. This is a finite set. Using $(*)$ for R', find a vertex $v' \in V'$ outside the image of φ (which is a finite set), so that v' is adjacent in R' to all the vertices in $\varphi(U)$ but to no other vertex in the image of φ. Put $\varphi(v) := v'$.

At even steps in the definition process we do the same thing with the roles of R and R' interchanged: we look at the first vertex v' in the enumeration of V' that does not yet lie in the image of φ, and set